Nicholas (Nico) Michael

AI Research & Development Lead | AI Software Developer | Solution Architect



1. Personal Information

Nationality: South African
Location: Pretoria, Gauteng, South Africa
Languages: English & Afrikaans

Contact Details

Email: nicomichael2018@yahoo.com
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YouTube DevOps Channel : https://www.youtube.com/@dvp7388

GitHub: https://github.com/nic0michael

AI Portfolio video: https://youtu.be/GyjAqyJZw_Y

Portfolio Development project using Outbox and VETO Design patterns:
https://github.com/nic0michael/Trans-Process-Platf/tree/master

2. Professional Summary

Senior Java Developer, Technical Lead and AI-Enabled Software Development Specialist with **26+ years of Java development** experience, and 5 years of Python coding experience.

Architecture Experience: 6 years across Solution and Infrastructure Architecture, designing enterprise, application and infrastructure solutions.

Delivered **56+ Spring Boot microservices into production at WesBank**. Worked across Banking, Finance, Healthcare, Insurance, Telecommunications, Education, Loyalty and Utility sectors.

Strong practical experience in Java, Spring Boot, Python, Kubernetes, Docker, Terraform, Azure AKS and CI/CD.

Development Best Practice Coach specializing in TDD, BDD, Clean Code, DevOps and AI-enabled software development.

Experienced in **applying AI development tools** and methodologies, including **Spec-Driven Developmen**t (SDD), **Context-Driven Development** (CDD)**, Iterative Context-Driven Development** (ICDD), and **multi-agent development**. Focused on enabling AI agents to develop software within defined requirements, coding standards, policies and quality controls

3. AI Skills - AI Development

AI-Assisted Development

Practical experience in development while evaluating and using GitHub Copilot, Claude Code, OpenAI Codex and Cursor.

Using AI tools for software development, problem-solving, code generation and technical analysis Including Reverse-Engineering Projects to Architecture Documentation .

Research and practical evaluation of local LLM development using Ollama, OpenCode and Google Gemma.

AI-Integrated Development

Researched and implemented Spec-Driven Development (SDD) using Microsoft GitHub Spec Kit.

Developed Iterative Context-Driven Development (ICDD), using AI-assisted brainstorming, requirement refinement and iterative quality reviews.

Applied Context-Driven Development (CDD) using Claude Code.

Defined structured specifications, requirements, acceptance criteria and coding standards to guide AI coding agents.

Applied human review and quality validation to AI-generated code.

Agentic AI Development

Evaluated single-agent and multi-agent development approaches.

Developed the Claude Code Multi-Agent project using Python FastAPI, CORS and Angular.

Researched agent coordination, task delegation, context management and policy-driven development.

Published open-source projects, including Claude Code Multi Agent With Policies.

AI Governance and Quality

Developed corporate AI policy files and coding governance guidelines for AI-generated code.

Defined development guardrails to constrain AI agents to approved requirements, standards and acceptance criteria.

Applied TDD, BDD, Clean Code, SonarQube and human peer reviews as part of software quality practices.

Evaluated AI development platforms using a common full-stack application and recommended Claude Code and Codex based on practical results.

4. Architecture Experience – AI and Cloud

Six years of architecture experience across Infrastructure and Solution Architecture, including AI architecture, agent-oriented design, cloud-native architecture and enterprise solution design.

AI Architecture

Researched AI Agentic Systems and Agent-Oriented Design.

Evaluated single-agent and multi-agent architectures.

Developed architectural approaches and governance guidelines for AI-assisted and agentic software development.

Created and maintained the AHAA methodology, drawing on C4, TOGAF, Zachman and IBM Global Services methodologies.

Developed an architecture methodology for Vertice:

- TOGAF – What to do?
- IBM GS Method – How to do it?
- C4 – What to draw?

Cloud and Infrastructure Architecture

Designed and implemented a High Availability K3S Kubernetes cluster spanning Germany, England and Johannesburg.

Led the integration of the Billing Switch application into Azure AKS Kubernetes.

Designed infrastructure and application architectures across financial services, telecommunications, utilities, information technology and healthcare environments.

Applied Docker, Kubernetes, K3S, Helm and Terraform to containerised and cloud-native deployments.

Practical cloud experience across Azure, AWS, GCP and Oracle Cloud.

Infrastructure architecture experience at IBM and ESKOM.

Solution architecture experience at Cell C, FleetDomain and Vertice.

5. Core Technical Competencies

- Software Engineering: **Java**, **Spring Boot, REST** APIs, JPA, Full Stack Development
- Front-end Development: **Angular 20, Cypress**(similar to Playwright), Jasmine
- Technical Leadership: **Team Leadership, Mentoring**, Training, Development Best Practices
- DevOps & Cloud: **Kubernetes, Docker, Helm, Jenkins, GitLab CI/CD, Terraform**
- Cloud Platforms: **Azure AKS, AWS, GCP, Oracle Cloud**
- Architecture: Solution Architecture, Infrastructure Architecture, Enterprise Architecture
- Cloud-Native: **Kubernetes, K3S, AKS, Docker, Helm,** Cloud-Native Architecture
- Software Quality: TDD, BDD, **JUnit, Mockito**, **SonarQube**, Clean Code
- REST: Microservices **Quarkus 3.8, Spring Boot**, and Python
- Database: **MMSQL, PostgreSQL, Oracle 11g to 19c, MySQL**, DB2
- JMS Queues: **RabbitMQ, Kafka**, IBM MQ series
- AI-Assisted Development: Spec Driven Development SDD, Context Driven Development CDD, Prompt Engineering, Multi-Agent Development
- AI Tools: **GitHub Copilot, Claude Code, Codex, Cursor**
- Process & Architecture: **UML, C4, TOGAF, IBM GS Method**
- SASS: **The Billing switch** was delivered as a SASS **in Azure AKS Kubernetes**

6. DevOps & Cloud Engineering Experience

Practical experience across DevOps, cloud infrastructure, automation and containerization.

- 2 years **Azure** experience including **Azure AKS Kubernetes** and CI/CD Pipelines.
- 1 year **GCP** experience developing a Notification App using Firebase in GCP.
- 1 year practical **AWS** experience building infrastructure for Udemy training.
- 4 years **Oracle Cloud** provisioning **Rocky Linux** servers
- 8 years **Docker** experience.
- 7 years **Kubernetes, including Azure AKS Kubernetes** and Cloud-Native platform experience.
- Applied **Terraform** for repeatable Infrastructure as Code deployments.
- Developed PowerShell and Linux shell automation for development, testing, analysis and deployment.
- Worked with **Docker Compose and multi-container environments**.
- Practical Kubernetes experience covering K3S, workloads, nodes, namespaces, pods, services and troubleshooting.
- Experience with Proxmox virtualised infrastructure.
- Experience with **SonarQube** for automated code-quality and security analysis.
- Practical troubleshooting of networking, ports, DNS, certificates, containers, virtual machines and deployment failures.
- Applied **DevSecOps** principles within the software delivery lifecycle.

7. Key Achievements

- Delivered **56 Spring Boot microservices** into production at WesBank.
- Designed and implemented a **High Availability K3S Kubernetes cluster** spanning Germany, England and Johannesburg.
- **Developed a Python-based K3S Kubernetes Cluster Generator** to automate Kubernetes cluster configuration and generation.
- Used Helm Charts to automate application deployment and management across Kubernetes.
- Created technical demonstrations **integrating Terraform, Kubernetes, Helm and Proxmox.**
- Introduced TDD, BDD and DevOps best practices across development teams.
- Developed our Agentic **Architecture and Development Methodology**.
- **Lead the integration of the Billing Switch** application into **Azure** AKS Kubernetes.
- Published a Microservices White Paper.
- Published 50 Open Source projects on GitHub, including the K3S **Kubernetes Auto Cluster Generator** and **Claude Code Multi Agent projects** with Corporate Policies.
- **Provided technical training** in Java, Kubernetes, DevOps and AI-Assisted Development.

8. Selected Technical Projects

AI Agentic Development Project – Technical Debt Reporting

Project Overview

This project was developed to **provide senior management with daily visibility of software technical debt**, including security vulnerabilities, bugs and code smells, and the estimated effort and cost required to remediate them

- Developed an AI-agentic application using **Claude Code, Java Spring Boot, Angular and PostgreSQL database**.
- Integrated with the S**onarQube Cloud API** to generate daily technical debt reports for senior management.
- Analyzed code submitted to **Azure DevOps Repositories**.
- Reported estimated remediation effort for:
    a. Security vulnerabilities.
    b. Software bugs.
    c. Code smells.
- Consolidated remediation estimates to help senior management budget and plan the cost of addressing technical debt.
- Used **Claude Opus as the orchestrator** and multiple **Claude Sonnet sub-agents**.
- Governed AI agents using corporate policies and coding standards defined in ***AGENTS.md**.

The Problem It Solved

When developers started using AI-agentic development, they were not consistently governing AI agents through **AGENTS.md** files or specifying the need for human-readable, maintainable code.
This resulted in **significant technical debt**, particularly a high volume of code smells.
I was asked to develop the application so senior management could monitor technical debt and track remediation progress.

How I Used AI

- Used **Claude Code** as the primary AI development environment.
- Implemented a multi-agent approach with **Claude Opus as orchestrator** and multiple **Claude Sonnet sub-agents** handling delegated development tasks.
- Guided agents using detailed requirements, corporate coding standards and policies defined in **AGENTS.md**.
- Used AI to accelerate development, delegate tasks in parallel and reduce manual coding effort.

Impact

As developers adopted **Claude Code and multi-agent development**, progress in addressing technical debt became visible through daily reporting. However, token consumption introduced a significant cost challenge.

One developer was using **Claude Opus for all agents**, resulting in high costs.
I recommended my strategy of using **Opus as orchestrator** and **Sonnet for sub-agent**s.
The developer adopted this approach and **reduced token costs** while retaining the benefits of **multi-agent development**.

Finally the app started revealing to senior management that the Technical debt began to reduce significantly

WesBank: Credit Scoring System 2018

Role: Senior Java Developer Team Lead

I was hired to deliver a Credit Scoring System, I **analyzed requirements** and designed the solution using UML. Designed a **non-synchronous, parallel-processing architecture** using RabbitMQ across a 4-node cluster, with Oracle for credit-score results and a micro-service for end-user access. Researched and prototyped a Drools rules engine to manage extensive business rules. Led implementation through to Production;
The scoring service code remained unchanged for 3 years

IBM: System Architecture for GJMC Metro Council of Johannesburg 2001

Role: System Architect

Took responsibility for the software and networking architecture across the GJMC Metro Council of Johannesburg. The requirement was to develop a complete Systems Architecture covering all customer regions, each with its own existing systems and requirements.

I visited each region to **identify and understand the existing systems** and infrastructure. I then produced a complete set of Architecture artefacts for each region, together with the overall GJMC Systems Architecture and the supporting documentation. This provided a consolidated architectural view of a large, geographically distributed environment

IBM: Architecture deliver for StanLib development project 2002

IBM had been contracted by StanLib to deliver an Insurance System **based on IBM's Insurance UML Model**, with development and implementation to be performed by a subcontractor. My responsibility was to ensure that the proposed system had the required infrastructure architecture to support its development and implementation.

I analyzed the proposed system and created the complete Infrastructure Architecture artefacts and supporting documentation required by the subcontractor. My colleague was responsible for the Solution Architecture; together we delivered the complete solution, including the required infrastructure, to the subcontractor for development and implementation.

Vertice: Dockerizing the Billing Switch Project taking it into the Azure AKS Kubernetes cluster 2025

This was done in 5 phases:

1. **Dockerized the Billing Switch** so new hires could easily install the required applications on their laptops.

2.**Taught Kubernetes to the team** performing the Azure AKS POC, including the Team Leader and DevOps team.

3. **Part of POC** team, **creating the Docker images** and **Docker containers** Kubernetes manifests.

4. I **created the Docker Containers** in our Azure DevOps Repository

5. Worked with DevOps, we created the pipelines and deploy the Billing Switch to the Development environment.

6. **I resolved Issues with the Kubernetes Manifests,** then deployed the Billing Switch to UAT and ultimately to Production.

9. Work History

Vertice

AI Research & Development Lead | AI Software Developer | Solution Architect | Aug 2025 - current

Focus: AI Research & Development, Agentic AI Software Development, AI-Assisted Development Methodologies, AI Governance, Cloud and Solution Architecture.

- Led research into AI-assisted software development, evaluating GitHub Copilot, Claude Code, Codex and Cursor.
- Researched and evaluated Spec-Driven Development (SDD) using Microsoft GitHub Spec Kit, Context-Driven Development (CDD) and agentic development methodologies.
- Investigated single-agent and multi-agent AI development approaches, including agent coordination, task delegation and policy-driven development.
- Applied AI development tools to build applications, including a Tech Radar providing management with technology adoption information.
- Developed a technical reporting application providing senior management with daily visibility of code quality, security issues and technical debt.
- Developed applications providing project managers with Git repository status and business analysts with development task status.
- Developed corporate AI policy frameworks and coding governance guidelines to establish standards and guardrails for AI-generated code.
- Created and maintained the AHAA **Agentic Architectural & Development Methodology**, drawing on C4, TOGAF, and IBM Global Services methodologies.
- Maintained the Architecture Intellectual Capital repository in SharePoint.
- Managed the Architectural Decision Register (ADR) process and repository.
- Maintained the **System Architecture Review (SAR) framework and supporting documentation**..

Tech Stack: **GitLab**, Azure DevOps Repo, Azure AKS Kubernetes, GitHub Copilot, GitHub Spec Kit, Claude Code, Cursor, Codex, VS Code, IntelliJ, Java, Python, Angular, Spring Boot **Python, Bash, Power Shell.**

Vertice

Senior Java Developer | Full Stack Developer | Best Practice Coach | Aug 2024 – Aug 2025

- Developed application code and provided technical training to development teams.
- Led the Dockerization of the Billing Switch.
- Mentored team members in Docker adoption.
- Participated in the Azure AKS Kubernetes POC.
- Guided the team through its first AKS Kubernetes adoption and integration.
- Provided Kubernetes and cloud-native development training.

- Introduced and delivered Helm training to the DevOps team.
- Trained development teams in TDD.
- Introduced and trained teams on Spec-Driven Development.
- Taught Context-Driven Development using Claude Code.
- Introduced advanced brainstorming techniques to improve SDD and CDD outcomes.

Tech Stack: Java, Spring Boot, Docker, Dockge, Kubernetes, K3S, Azure AKS, Helm, **GitLab**, Azure DevOps, **Python, Bash, Power Shell, Prometheus**

LoyaltyPlus

Senior DevOps Consultant / **Senior Developer** | Jun 2023 – Jul 2024

- Designed and **implemented Disaster Recovery** and backup strategy for customer WordPress platforms using Ansible, Docker, HAProxy and Semaphore.
- Built and automated **High Availability K3S Kubernetes cluster** across Germany, England and South Africa.
- Developed **infrastructure automation using Ansible, Terraform, Proxmo**x, Cockpit and Rocky Linux.
- Developed a distributed messaging platform using Camel Quarkus, Kafka, Angular and REST APIs supporting SMS, Email, WhatsApp, Push Notifications and In-App Notifications.
- Implemented Firebase integration for mobile notification services.
- Designed security monitoring and vulnerability scanning using Metasploit and StarRocks.
- Evaluated and implemented Zero Trust networking using NetBird.

Tech Stack: Java, Spring Boot, **Camel, Quarkus, Kafka, Angular, Docker**, K3S **Kubernetes, Helm, Ansible, Terraform,** Proxmox, **HAProxy, Semaphore, NetBird, Metasploi**t, **Python**, **BASH**,StarRocks, **Google GCP, Firebase**, REST APIs, Rocky Linux Servers.

Advance / MTN

Senior Software Engineer | Kubernetes Specialist | Technical Trainer | Jun 2022 – Jun 2023

- Developed solutions for the MTN USSD platform.
- Designed and implemented **High Availability K3S Kubernetes clusters** on Red Hat infrastructure.
- Created Kubernetes manifests and Helm Charts for Zookeeper, Elasticsearch, Kibana, Prometheus, PostgreSQL and MongoDB.
- Led Kubernetes adoption and deployment practices across development teams.
- Delivered training in TDD, JUnit, Mockito, Kubernetes, Helm and cloud-native deployment practices.

Tech Stack: Java, Spring Boot, **Kubernetes**, K3S, **Helm**, **Docker**, Elasticsearch, Kibana, **Prometheus**, PostgreSQL, MongoDB, Zookeeper, JUnit, Mockito, **Jenkins**, **Terraform, Bash, Python**

DVT

Senior Java Developer Team Lead | Feb 2017 – Jun 2022

2022

- Led Spring Boot Micro-service development initiatives.
- Developed Java training curricula and technical enablement programs.

Tech Stack: Java, Spring Boot Micro-services,Hibernate,GitLab, TDD, Mockito.

Curro Schools | 2021

- Served as Software Development Best Practices Coach and Senior Java Developer.
- Delivered Proofs of Concept covering TDD, BDD, **Docker**, **SonarQube**, Portainer, RabbitMQ, Postman and Micro-service architecture.
- Trained development and QA teams in automated testing and engineering best practices.

Tech Stack: Java, Groovy, Grails, Spring Boot Micro-services,**OAuth**, TDD, **SonarQube**

WesBank | 2018 feb– 2020 Jun

- Delivered 56+ Spring Boot microservices into production.
- Led adoption of TDD, BDD, **SonarQube**, **Bitbucket** and Confluence.
- Designed a KYC Rules Engine using Drools and RabbitMQ.
- Architected integration services for LDAP, App Notifications, CT02, CIS, Actimize and Documentum.
- Produced solution designs, architectural artifacts and technical documentation.
- Received **Kubernetes training** from WesBank
- Completed **AWS** training through Udemy and online practical for Just over 1 Year .

Tech Stack: Java, **Spring Boot**, **SOAP, RabbitMQ, Docker**, Docker Compose, Portainer, **SonarQube**, **Postman**, JUnit, Mockito, TDD, BDD, **Drools**, **Kubernetes**, Elasticsearch, Logstash, Kibana, LDAP, IBM ODM, **Nexis**, **Bash, Java API**

Sanlam | 2020

- Developed Spring Boot microservices integrated with IBM ODM Business Rules Management Engine.
- Used IBM Rule Designer to develop process flow diagrams.

Tech Stack: Java, **Spring Boot**, JUnit, TDD, **IBM ODM**, **GitLab**

Momentum | 2017 feb – 2018 feb

- Was the Full-stack Developer Team Lead
- Responsible for architecture, TDD adoption, code quality.
- Worked with **Docker**, **Docker Swarm** and **Ansible** automation

Tech Stack: Java, **Angular 2,Typescript, Bootstrap**, **Docker**, Docker Swarm, Postman, JUnit, Mockito, TDD, IBM ODM, **Ansible, GitLab**

FleetDomain

Solutions Architect | Infrastructure Architect | Senior Java Developer | May 2016 – Jan 2017

- Implemented enterprise security policies, disaster recovery, backup automation and UAT environments.
- Designed High Availability production architecture and hot backup solutions.
- Reverse-engineered and optimised Groovy/Grails enterprise applications.
- Modernised deployment architecture and server provisioning processes.

Tech Stack: Java, Groovy, Grails, **Ubuntu Server**, Windows Server, **Linux Administration,** Disaster Recovery, Security Architecture, Backup Automation, **Bash, GIT**

Standard Bank

Senior Java Developer | Jul 2015 – May 2016

- Developed enterprise OTP authentication services for Africa and Offshore Banking platforms.
- Delivered JEE8 interceptor framework and Vasco OTP integration.
- Applied TDD and BDD using JUnit, Mockito, WireMock, JBehave and Serenity.

Tech Stack: Java, JEE8, **SOAP**, **Maven**, JUnit, Mockito, WireMock, JBehave, Serenity BDD, SOAP, Web Services. **Centos Linux**, Puppet, **Java API**.

Cell C

Java Solution Architect | Infrastructure Architect | Senior Java Developer | Mar 2014 – Jul 2015

- Redesigned enterprise systems for scalability and performance.
- Changed the Architecture of Badly performing system
- Analyzed the Siemens Nokia Synix architecture Produced UML models.
- Optimized the Complete System Design.
- Made substantial improvement to performance

Tech Stack: Java, UML, Enterprise Architect, EJB, Oracle SQL, **Centos Linux**, Percona MySQL, JUnit,TDD, Javascript, **Bash**.

Signio

Senior Java Developer | Nov 2009 – Feb 2014

- Developed Standard Bank integration gateways and insurance platform integrations.
- Delivered Jasper Reports solutions and Groovy/Grails applications.
- Developed a project for the VAF5 System to send requests to the NATIS gateway
- Delivered new gateways to the Insurance companies

Tech Stack: Java, EJB3, SOAP, **Centos Linux**, Percona MySQL, Groovy, Grails, **Jasper Reports, Bash**

Paracon

ESKOM

Infrastructure Architect | Senior Java Developer | Apr 2007 – Jul 2009

- As Infrastructure Architect, I delivered the Architectural Artifacts for the Online Vending System and got approval from their Architecture Review Committee
- Installed and configured enterprise integration platforms and middle-ware.
- Developed BPEL Business Processes for SOAP integration connecting SAP using Sun Java CAPS.
- Worked as **Infrastructure Architect** for 2 years.
- We used the **VETO and VETRO SOAP Integration Patterns**

Swot Team

- Participated in an ESKOM SWOT team as Infrastructure Architect to resolve a severely under-performing system.
- Identified performance issues with a **Java CAPS Correlator** processing approximately 10 million message pairs.
- Collaborated with the team to challenge the existing design and develop an alternative solution.
- Proposed two indexed database tables with a join to match the two message types.
- **Achieved approximately 100 times faster performance**, helping move the system towards Production.

Tech Stack: Java, **SOAP, BPEL, LDAP**, Sun Java CAPS, Infrastructure Architecture, **Ubuntu Linux Server**, **Solaris Unix, Bash, VMware ESX Server, SUN VM Server**

ABSA Bank

Senior Java Developer | Jan 2006 – Mar 2007

- Developed Home Loans portal solutions using JSR168 Portlets and JSF.
- Participated in delivery of the production Home Loans platform.

Tech Stack: Java, JSR168 Portlets, JSF, **IBM AIX Unix**

Standard Bank

Senior Java Developer | Jul 2002 – Dec 2005

- Developed Home Loans portal solutions using JSR168 Portlets and JSF.
- Participated in delivery of the production Home Loans platform.

Tech Stack: Java, Java EJB2,Vignette CMS, **Solaris Unix**

IBM Global Services

Infrastructure Architect | Java Developer | Feb 1998 – Jun 2002

- Developed components of the National Online Casino Gambling Monitoring System.
- Worked as Infrastructure Architect.
- Delivered **Infrastructure Architecture for StanLib project**
- Created Architectural artifacts for the GJMC project for Johannesburg Metro Council
- I conducted the **CMM Maturity Rating of Rubico** for the World Bank
- Was part of the IBM **GS Method Methodology** training team of IBM

Tech Stack: Java, Java EJB, Java J2EE, RUP, Visual Age for Java, UML Sybase SQL server, IBM DB2, Windows NT Server, O/S2

10. Education, Professional Training & Qualifications

National Higher Diploma – Production Management

Tshwane University of Technology | Graduation 1989

National Diploma – Electrical Engineering L/C

University of Johannesburg | Graduation 1980

Matric

TED | 1975

Professional Training:

SQL,Java, EJB JEE, UML, C++, Kubernetes, Docker, Jenkins, Terraform, Azure, AWS, Metasploit and Agile Methodologies.

Training providers include **IBM, Damelin and Udemy**

11. Publications, Open Source & Community Contributions

Published Microservices White Paper on Test Driven Development.

- https://www.dvt.co.za/white-paper-building-cloud-ready-microservices-using-industry-best-practices

Published 50 Open Source projects on GitHub.

- Created the K3S Kubernetes Auto Cluster Generator
- The Claude Code Multi Agent Development project With Policies.
- Technical training in Java, Kubernetes, DevOps and AI-Assisted Development.

Published Videos in DevOps YouTube channel

- Videos covering Java, Enterprise Java, **Quarkus**, TDD, **Docker, Kubernetes**, Portainer, **Proxmox**, **Terraform**/OpenTofu, **Helm, Jenkins, Ansible**, AI Agentic Development and N8N.

12. Professional Memberships

- SAIETE – South African Institute for Technician Engineers
- SAIEE – South African Institute for Electrical Engineers
- JUGSA – Java User Group of South Africa
- GLUG – Gauteng Linux User Group

References

Professional and personal references available upon request.